

Profitability & Sales Optimization

Data warehouse → Python analytics → Power BI insights





Tools & Environment



Data

SQL Server, SSMS, Data Warehouse (Bronze → Silver → Gold)



Analytics

Python (Pandas, NumPy, Matplotlib, Seaborn), Jupyter



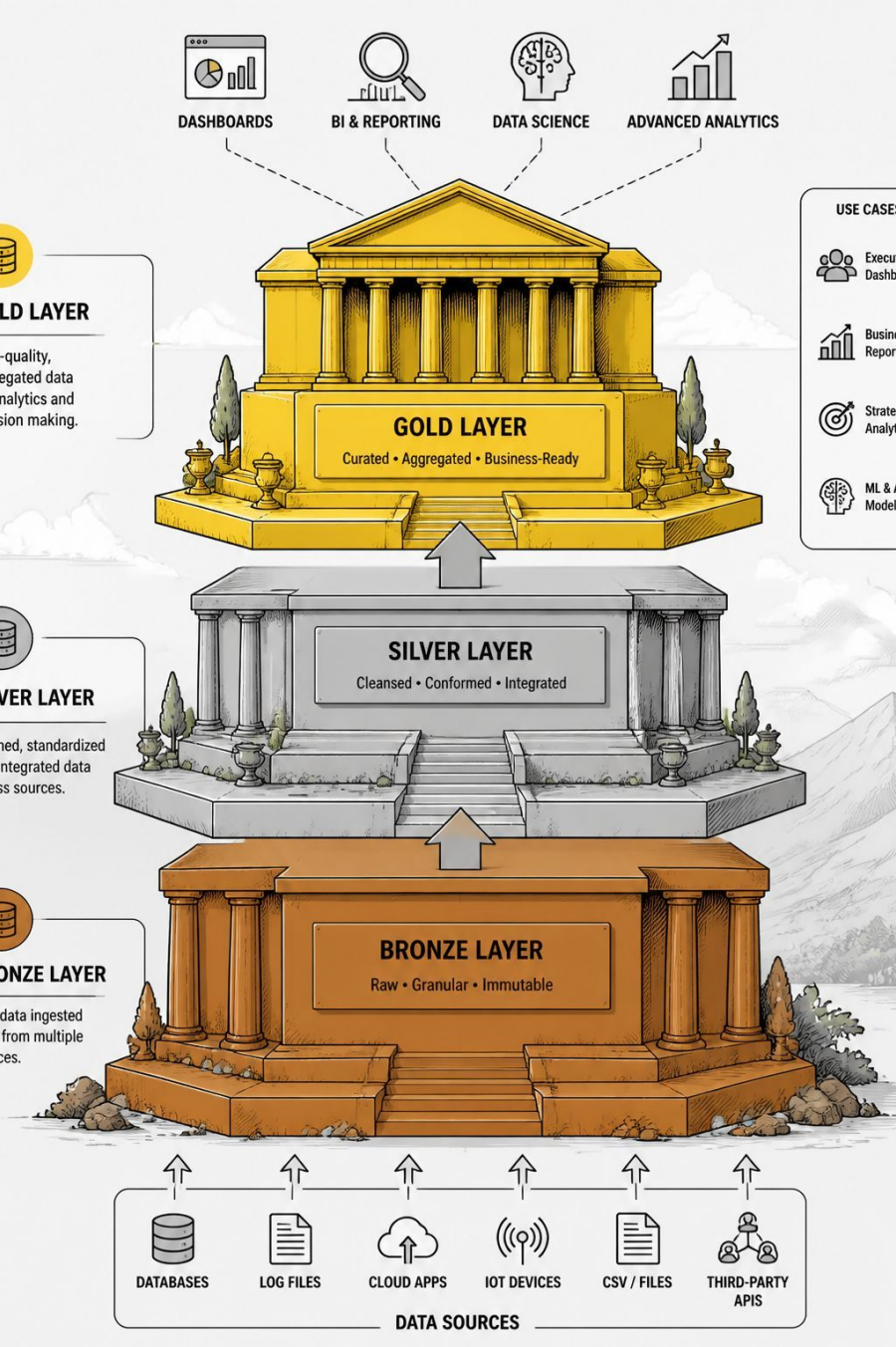
Reporting

Power BI dashboards, DAX measures, PowerPoint



Ops & Docs

GitHub, Notion, Draw.io, Excel, Word, Power Point, Gamma.



Data Engineering

Bronze

Raw tables; DDL + load procedures

Silver

Cleaned, standardized tables

Gold

Analytical views for reporting

Project Objective

Business Goal

Identify most & least profitable product lines and regions to optimize marketing and pricing



Product & Category Questions



Revenue Illusions

Top 20% revenue
but bottom 50%
margin?



Hidden Gems

High-margin, low-
volume items for
marketing lift



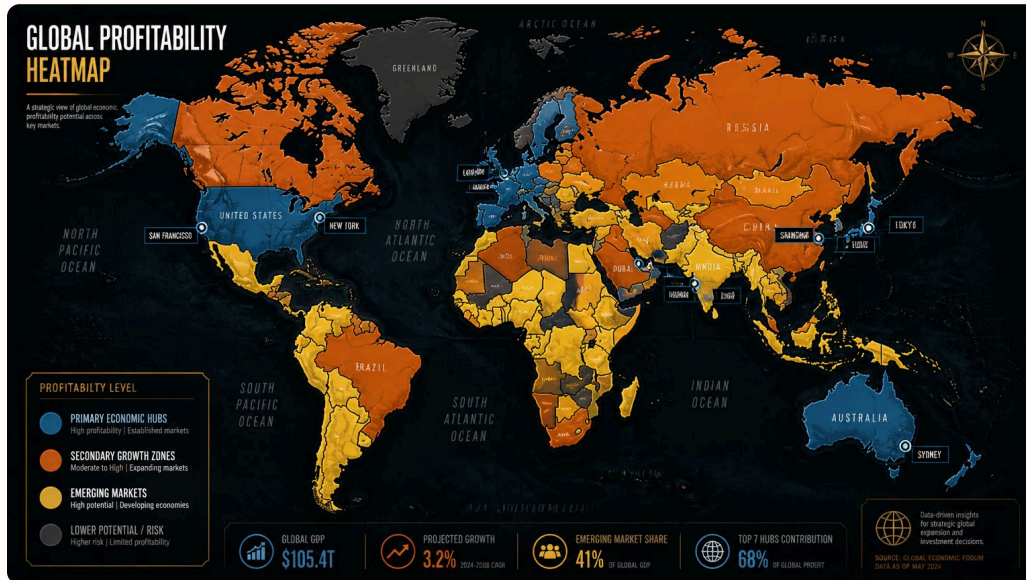
Cost vs Margin

Correlation
between
product_cost and
profit margin



Maintenance Impact

Does
maintenance flag
reduce lifetime
margin?



Regional & Geographic Insights

- Rank countries by avg profit margin per order
- Detect regional pricing inefficiencies by category
- Map top 20% profitable customers by country

Customer Demographics & Behavior



Top Customers

Profit contribution
of top 10%
customers



Revenue vs Profit

Are high-revenue
customers low-
profit?



Demographic Cohorts

Age, gender,
marital_status →
highest avg profit

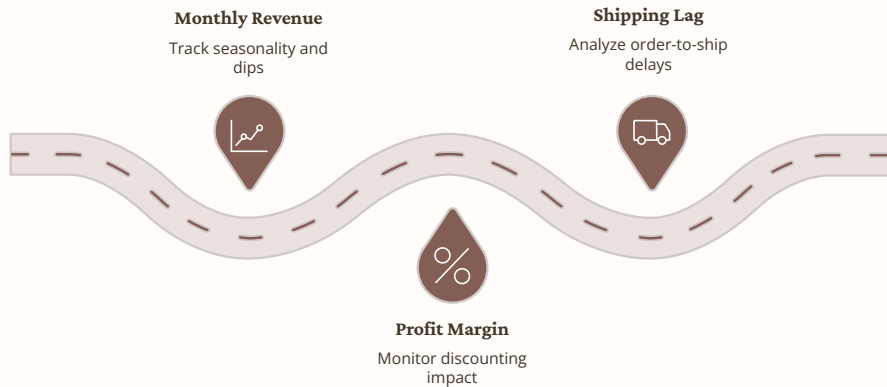


Order Frequency

Correlation: order count vs profit margin



Time Series & Ops Efficiency

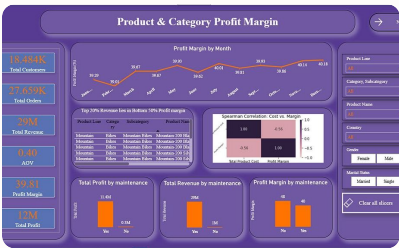


Seasonal dips: revenue months vs margin (discounting effect)

Shipping lag: order_date → shipping_date by product line

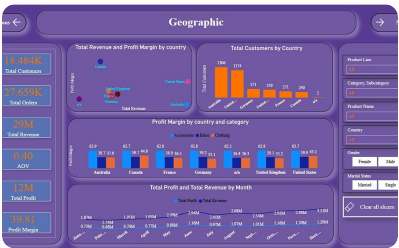
Use results to schedule campaigns and fix bottlenecks

Dashboards & Measures



Product & Category Page

Product profitability, margins, rankings



Geographic Page

Country ranking by avg margin



Customer Page

Top customers, cohorts, order behavior

Implementation & Artifacts



Code & Analysis

Python notebooks, EDA scripts, correlation & pie charts



SQL Assets

DDL, load procs, EDA reports, ranking procedures



Repository

GitHub: Profitability-and-Sales-optimization (project + dashboard)



Deliverables

Power BI dashboards, PowerPoint, documentation (Word, Notion)

📄 Theme accent: #481c9eff